

APPLIED MACHINE LEARNING TECHNIQUES · MODULE 4

Unsupervised Learning, Made Simple

Clustering · Association Rules · Evaluation Metrics — explained with everyday analogies, worked math, and zero jargon.



K-Means · GMM



Market Baskets



Cluster Quality

What We'll Cover Today

- ▶ What unsupervised learning is (and isn't)
- ▶ How it differs from supervised learning
- ▶ Real-world uses: marketing, healthcare, security
- ▶ Anomaly detection: point, contextual, collective
- ▶ Customer segmentation in business
- ▶ K-Means clustering + the Elbow Method
- ▶ Hierarchical clustering & dendrograms

- ▶ Gaussian Mixture Models & soft clustering
- ▶ Association Rule Learning
- ▶ Market basket analysis & rule mining
- ▶ Support, Confidence, and Lift — by hand
- ▶ Apriori, FP-Growth, and Eclat algorithms
- ▶ Cosine similarity, step by step
- ▶ Judging clusters: Silhouette & Davies-Bouldin

GOALS

By the End, You Will Be Able To...



Tell apart supervised and unsupervised learning — instantly, with examples.



Spot where unsupervised learning quietly runs the real world around you.



Cluster data using K-Means, Hierarchical methods, and Gaussian Mixture Models.



Mine “bought-together” rules with Apriori, FP-Growth, and Eclat.



Score cluster quality using Silhouette Score and the Davies-Bouldin Index.

PART 1

01

Introduction to Unsupervised Learning

- The core idea — learning without an answer key
- Everyday analogies: jigsaw puzzles, fruit baskets, and bikes
- The process flow, and the three big approaches
- Supervised vs. unsupervised — side by side

What Is Unsupervised Learning?

Unsupervised learning is machine learning on data that carries **no labels and no right answers**. The model isn't told what to find — it discovers hidden patterns, groups, and relationships on its own.



Analogy: a jigsaw with no box photo

You've lost the box, so nobody can tell you what the final picture looks like. You still make progress: sky-blue pieces in one pile, straight edges in another, green textures in a third. You grouped by similarity — exactly what unsupervised learning does.



Contrast: studying with an answer key

Supervised learning is like practicing with solved papers — every question comes with the correct answer, and you learn the mapping. Unsupervised learning is handed only the questions and asked: “what structure do you see?”

INTUITION CHECK

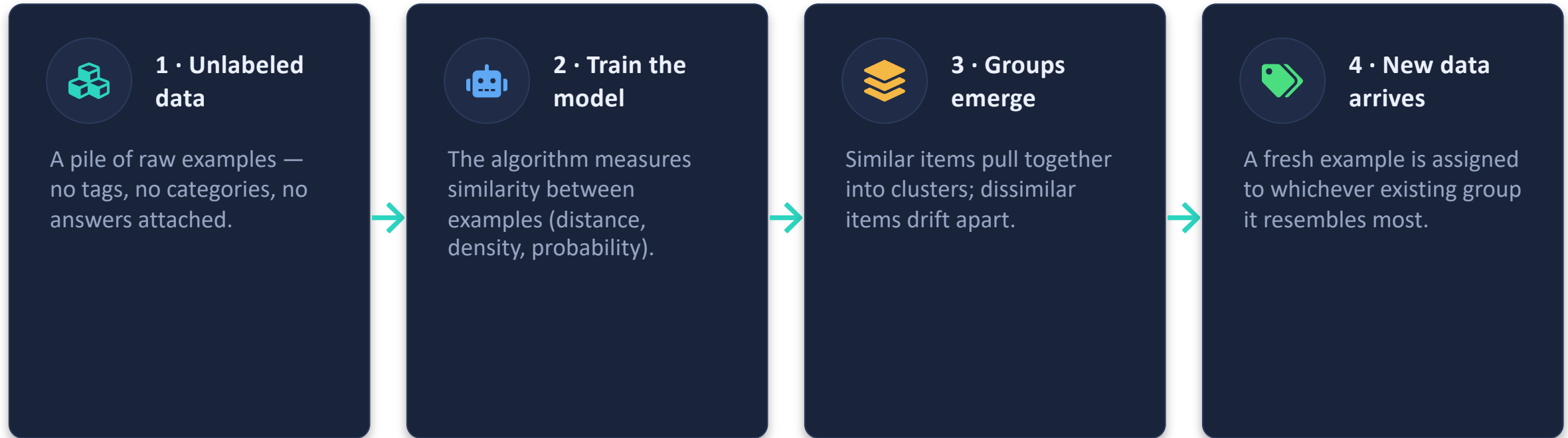
Can You Guess the Fruit From Features Alone?

No labels are given — only color, shape, and size. Your brain clusters them anyway. That instinct is the algorithm.

Color	Shape	Size	Your brain says...
Yellow	Long, curved cylinder	Big	Banana
Red	Round, dip on top	Big	Apple
Green / purple	Small ovals in a bunch	Small	Grapes

Key point: you never saw a label saying “banana”. You formed groups purely from shared features — and a name for each group came *after* the grouping. Unsupervised models work in exactly this order: cluster first, interpret later.

The Unsupervised Learning Process Flow



Note: “training data” here just means a collection of information **without any labels** — the model is never shown a correct answer at any point.

Sorting Vehicles Without Being Told What They Are

- 1 Feed in a mixed set of two-wheelers — bicycles, scooters, sports bikes — described only by features: size, top speed, engine, color.
- 2 The system notices natural groupings: small + pedal-powered, medium + step-through, large + high speed — and forms categories on its own.
- 3 Show it a brand-new vehicle: it compares features and drops it into the most similar existing group.

The groups depend entirely on the attributes you choose. Feed in color → you get red vs. green piles. Feed in speed → racing vs. commuter piles. Same data, different lens, different clusters. Feature choice is the steering wheel of unsupervised learning.

Three Families of Unsupervised Learning



Clustering

Group similar things together.

Like sorting a laundry basket: whites, colors, delicates — no one labeled the clothes, the similarity did the work.

Examples: K-Means, GMM, Hierarchical.



Association Rules

Find what goes together.

Like noticing chai is almost always ordered with biscuits — a “if A, then probably B” pattern.

Examples: Apriori, FP-Growth, Eclat.



Dimensionality Reduction

Compress many features into few.

Like a movie trailer: 2 minutes that keep the story of 2 hours. Less data, same essence.

Examples: PCA, t-SNE, Autoencoders.

This module focuses on the first two — clustering and association rules.

Supervised vs. Unsupervised Learning

Aspect	Supervised	Unsupervised
Data	Labeled — every example has an answer	Unlabeled — raw examples only
Goal	Predict outcomes (classify / regress)	Discover structure (clusters / rules)
Human effort	Heavy — someone must label the data	Light — no labeling needed
Algorithms	Linear Regression, SVM, Decision Trees	K-Means, PCA, Hierarchical, Apriori
Typical use	Spam detection, fraud prediction	Customer segments, topic discovery

One-line memory hook: supervised = exam with an answer key; unsupervised = detective work with no key, only clues.

Same Data, Different Question

Supervised

You get pairs: $(x_1, y_1), (x_2, y_2), \dots$

Learn a function $f: X \rightarrow y$ that maps inputs to known answers.

Unsupervised

You get only inputs: $x_1, x_2, \dots, x_n$

Learn the **structure of X itself** — groups, co-occurrence rules, or a compressed form.

Concrete example — same spreadsheet of customers:

Supervised question: “Given past churners (labels), will THIS customer churn?”

Unsupervised question: “With no labels at all, what natural customer groups exist in here?”

PART 2

02

Where It's Used in the Real World

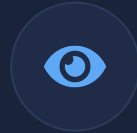
- Five everyday applications you already interact with
- Deep dives: healthcare, marketing, cybersecurity
- The honest list of challenges

It's Already All Around You



News grouping

Google News bundles hundreds of articles about the same event — nobody labels them; similarity of words does it.



Computer vision

Grouping visually similar images or regions — e.g., photo apps clustering faces without knowing names.



Medical imaging

Separating tissue types in MRI/CT scans by pixel patterns, helping flag unusual regions.



Anomaly detection

Learning what “normal” looks like, then raising a flag when something deviates — fraud, intrusions, faults.



Recommendations

“Customers like you also bought...” — grouping users by behavior, not by declared identity.

USE CASE 1

Healthcare — Finding Patient Groups No One Defined

Hospitals cluster patients by patterns in vitals, lab results, and history — **without predefined risk labels** — and discover risk segments the staff never explicitly created.

Cluster found	Pattern (BP, cholesterol)	What doctors do with it
Group A	Low BP, low cholesterol	Routine annual checkups
Group B	Borderline values, drifting upward	Lifestyle program + 6-month review
Group C	High BP and high cholesterol together	Priority screening & medication review

The win: the “high-risk” label was never in the data — the cluster surfaced it, and humans named it afterwards.

USE CASE 2

Marketing — Customer Segmentation

Group customers by behavior — purchase frequency, average spend, browsing — **with no predefined segments**. The algorithm draws the boundaries; marketing names them.

Customer	Visits / month	Avg spend (₹)	Cluster → nickname
C1	13	3,100	Frequent big spenders
C2	2	420	Occasional shoppers
C3	7	1,100	Price-sensitive regulars
C4	14	3,250	Frequent big spenders

Payoff: send VIP perks to big spenders, discount coupons to the price-sensitive — same budget, far better conversion.

USE CASE 3

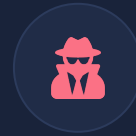
Cybersecurity — Catching Attacks Nobody Has Seen

The model learns what **normal** user and network behavior looks like. Anything that lands far from that normal cloud gets flagged — **no labeled attack examples required.**



Normal looks like...

Sessions of 80–120 seconds moving 400–600 KB, clustered tightly together during office hours. Dense, boring, predictable — exactly what we want.



Suspicious looks like...

A 3 a.m. session moving 950 KB in 30 seconds, sitting far outside the dense cloud. The model has never seen this attack before — distance alone betrays it.

Why this matters: supervised systems only catch attacks they were trained on; unsupervised systems can catch attacks invented yesterday.

Five Challenges of Unsupervised Learning



Computational cost: Comparing everything with everything is expensive — big data makes it heavier.



Longer training time: No answer key means more exploration before the structure settles.



Risk of wrong results: With no ground truth, the model can group things confidently — and wrongly.



Humans must validate: Someone has to inspect clusters and decide if they actually mean anything.



Low transparency: It's often unclear why two items landed in the same cluster — explainability suffers.